

Loyola Marymount University

# CMSI 3300 Artificial Intelligence

Fall 2024

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# Announcements

- Grades of Classwork 1 and 2 are posted on Brightspace
- Solution of Classwork is also posted on Brightspace
- Classwork 3
  - Due this **Friday 9/27 @5 pm**
- Homework 1
  - Due Friday 10/4 @5 pm
  - Progress: Start on MazeAgent (Part 2)
- STEM Career Fair this Thursday 2-5 PM UHall

# This Lecture

- Bayesian Network
  - Exact Inference
  - Enumeration Inference
  - Inference Efficiency

# Last Lecture

- Inference Goal of BN
  - $BN.ask(Q, e) = P(Q|e)$
- How to compute $P(Q|e)$ given BN?
  - BN
    - Structure (i.e., DAG)
      - Independence/Conditional Independence
    - Semantics (i.e., CPTs)

How to determine it?

# Last Lecture

- d-Separation

| Triplet Type | Format | Rule (Blocked if …) |
|---|---|---|
| Fork | $X \leftarrow Z \rightarrow Y$ | Blocked if **Z is given** |
| Chain | $X \leftarrow Z \leftarrow Y$<br>$X \rightarrow Z \rightarrow Y$ | Blocked if **Z is given** |
| Sink | $X \rightarrow Z \leftarrow Y$ | Blocked if **neither Z nor its descendants are given** |

# Exact Inference in BN

- What is the goal of inference in BN?
  - $BN.ask(Q, e) = P(Q|e)$
- However, certain queries are easier to answer than others. Why?
  - Arbitrary queries of the format $P(Q\,|\,e)$ are not always immediately available from the BN.
- What are immediately available from the BN?
  - Structure (i.e., DAG) and Semantics (i.e., CPTs)

# Query Difficulty

- O(1) Answerable Queries
  - Can be answered in $O(1)$ **directly from** one of the network's **CPTs / parameters** (without needing to perform any computation).
- Example: $X \rightarrow Z \rightarrow Y$
  - CPTs: P(X), P(Z|X), P(Y|Z)
  - Query: P(Z=1|X=0)
    - Answerable in O(1) by looking up in CPT on Z

What are the CPTs?

# Query Difficulty

- O(|V|) Answerable Queries
  - **Linearly Answerable** O(|V|) Queries (for |V| variables) are those can be answered using **ONLY Markovian Factorization**.
  - A row in JPT that can be reconstructed from CPTs
- Example: $X \rightarrow Z \rightarrow Y$

What is the JPT given the CPTs ?

  - JPT: P(X, Z, Y) = P(X) P(Z|X) P(Y|Z)
  - Query: P(X=0, Z=1, Y=1)
    - = P(X=0) P(Z=1|X=0) P(Y=1|Z=1)

# Query Difficulty

- O(|V|$^2$) Answerable Queries
  - **Quadratic Answerable** O(|V|$^2$) Queries are those for which application of **d-separation**, with cost O(|V|) , need be applied at most $|V|$ times to reduce a query to a O(1) format.
- Example: $X \leftarrow Z \rightarrow Y$
  - Query: P(X|Z,Y)
  - d-Sep: P(X|Z,Y) =P(X|Z) → O(1) query by looking up CPT on X

# Example

- BN: $A \rightarrow B \rightarrow C$
- What is the difficulty of each query?
  - P(B|A) O(1)
  - P(C|A,B) $O(|V|^2)$
  - P(A|B) Cannot be obtained efficiently.
  - P(A,B,C) $O(|V|)$

How to compute it?

# (Other) Computable Queries

- Cannot be obtained **efficiently** using the known techniques.
- Need to be computed using rules of probability theory
  - Conditioning $P(Q|e) = \frac{P(Q,e)}{P(e)}$
  - Law of Total Probability $\mathrm{P}(\alpha) = \sum_{w \in M(\alpha)} P(w)$
  - Markovian Factorization

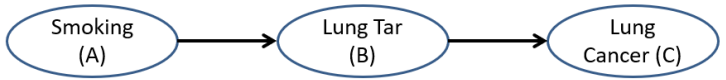
$$P(V_1, V_2, \dots) = \prod_{V_i \in \boldsymbol{V}} P(V_i | parents(V_i))$$

# Enumeration Inference

- **Enumeration Inference** is an inference algorithm that can compute any query via **sums-of-products of conditional probabilities** from the CPTs.
- Example
  - Query: P(C|A=1)

| A | Pr(A) |
|---|---|
| 0 | 0.4 |
| 1 | 0.6 |

| B | C | Pr(C \| B) |
|---|---|---|
| 0 | 0 | 0.5 |
| 0 | 1 | 0.5 |
| 1 | 0 | 0.7 |
| 1 | 1 | 0.3 |

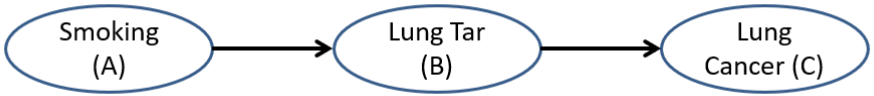



| A | B | Pr(B \| A) |
|---|---|---|
| 0 | 0 | 0.8 |
| 0 | 1 | 0.2 |
| 1 | 0 | 0.1 |
| 1 | 1 | 0.9 |

# Step 1: Identify Variable Types Q, e, Y

- **Query Variable** ($Q$): a single variable for which we are interested in computing the likelihood
- **Evidence Variables** ($e$): are variables that are observed at some value, and are held as evidence
- **Hidden Variables** ($Y$): are any variables in the network that are not part of the query
- What are the Q, e, Y in the example?
  - Q={C}, e={A=1}, Y={B}

Query: P(C|A=1)



# Steps 2-4: Compute P(Q|e)

$$P(Q|e) = \frac{P(Q,e)}{P(e)}$$

Step 4: Normalization

$$= \frac{1}{P(e)} \cdot P(Q,e)$$

$$= \frac{1}{P(e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

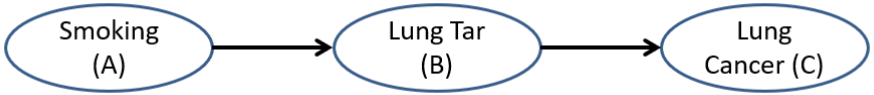
joint distribution: can be factored w/ MF

$$= \frac{1}{\sum_{q \in Q} P(Q,e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

Step 2: P(Q,e)

Step 3: P(e)

# Step 2.1: Compute P(Q,e)




Query: P(C|A=1)

$$P(Q,e) = \sum_{y \in Y} P(Q,e,y)$$

$$P(C, A=1) = \sum_b P(C, A=1, B=b)$$

Can we factor this?

$$P(C, A=1) = \sum_b P(C|B=b)P(B=b|A=1)P(A=1)$$

Blocked “Chain”

# Step 2.2: Optimize the Order

$$P(C, A = 1) = \sum_b P(C|B = b)P(B = b|A = 1)P(A = 1)$$

Can any factor be pulled outside of the summation?

It does not depend on b.

$$P(C, A = 1) = P(A = 1) \sum_b P(C|B = b)P(B = b|A = 1)$$

Are we ready to compute it?

# Step 2.3: Plugging In

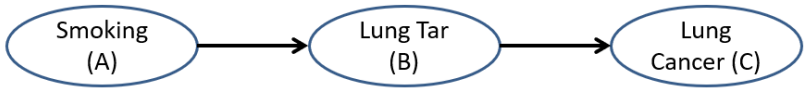
$$P(C, A = 1) = P(A = 1) \sum_b P(C|B = b)P(B = b|A = 1)$$

Let's try it…

1. $P(C = 0, A = 1)$
2. $P(C = 1, A = 1)$

| A | Pr(A) |
|---|---|
| 0 | 0.4 |
| 1 | 0.6 |

| B | C | Pr(C \| B) |
|---|---|---|
| 0 | 0 | 0.5 |
| 0 | 1 | 0.5 |
| 1 | 0 | 0.7 |
| 1 | 1 | 0.3 |




| A | B | Pr(B \| A) |
|---|---|---|
| 0 | 0 | 0.8 |
| 0 | 1 | 0.2 |
| 1 | 0 | 0.1 |
| 1 | 1 | 0.9 |

# Step 2.3: Plugging In

| $C$ | $P(C, A = 1)$ |
|---|---|
| $C = 0$ | $P(C = 0, A = 1) = P(A = 1) \sum_{b \in B} P(C = 0 \mid B = b) P(B = b \mid A = 1)$<br>$= P(A = 1)[P(C = 0 \mid B = 0) P(B = 0 \mid A = 1) + P(C = 0 \mid B = 1) P(B = 1 \mid A = 1)]$<br>$= 0.6 * [0.5 * 0.1 + 0.7 * 0.9]$<br>$= 0.408$ |
| $C = 1$ | $P(C = 1, A = 1) = P(A = 1) \sum_{b \in B} P(C = 1 \mid B = b) P(B = b \mid A = 1)$<br>$= P(A = 1)[P(C = 1 \mid B = 0) P(B = 0 \mid A = 1) + P(C = 1 \mid B = 1) P(B = 1 \mid A = 1)]$<br>$= 0.6 * [0.5 * 0.1 + 0.3 * 0.9]$<br>$= 0.192$ |

# Steps 2-4: Compute P(Q|e)

$$P(Q|e) = \frac{P(Q,e)}{P(e)}$$ Step 4: Normalization

$$= \frac{1}{P(e)} \cdot P(Q,e)$$

$$= \frac{1}{P(e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

$$= \frac{1}{\sum_{q \in Q} P(Q,e)} \cdot \sum_{y \in Y} P(Q,e,y)$$ Step 2: P(Q,e)

**Step 3: P(e)**

# Step 3: Compute P(e)

- From Step 2:
  - P(C=0, A=1) = 0.408
  - P(C=1, A=1) = 0.192
- Can we compute P(A) (or P(e))?
  - $P(e) = \sum_{q \in Q} P(Q, e)$
  - P(A=1) = P(C=0, A=1) + P(C=1, A=1) = 0.408+0.192= 0.6

# Step 4: Normalization P(Q|e) = P(Q,e)/P(e)

- From Steps 2 and 3:
  - P(C=0, A=1) = 0.408
  - P(C=1, A=1) = 0.192
  - P(A=1) = 0.6

| C | P(C\|A=1) |
|---|---|
| 0 | P(C=0, A=1)/P(A=1)=0.408/0.6=**0.68** |
| 1 | P(C=1, A=1)/P(A=1)=0.192/0.6=**0.32** |

# Inference Efficiency

- Efficiency Improvements
  - Ignore some CPTs
  - Example
    - P(B|A=a, E=e)
    - Which variables are irrelevant to the query?
      - {J,M}

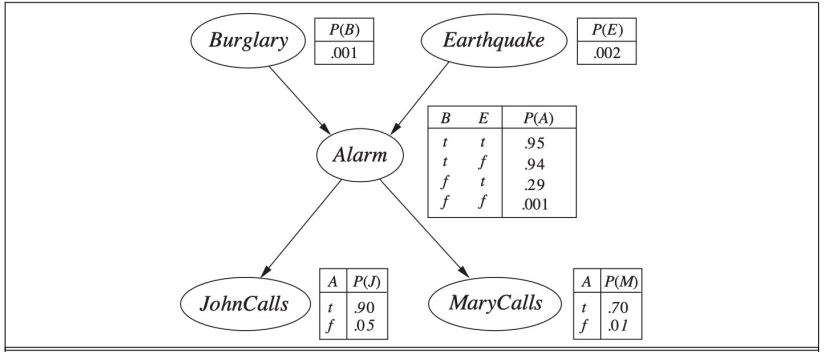



**Figure 14.2** A typical Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters $B$, $E$, $A$, $J$, and $M$ stand for *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, and *MaryCalls*, respectively.

# Simplification

- P(B | A=a, E=e)

$$\alpha = \frac{1}{P(A=a,E=e)}$$

$$P(B|A=a,E=e) = \alpha \sum_j \sum_m P(B)P(E=e)P(A=a|B,E=e)P(J=j|A=a)P(M=m|A=a)$$

$$= \alpha\, P(B)P(E=e)P(A=a|B,E=e) \sum_j P(J=j|A=a) \sum_m P(M=m|A=a)$$

=1

Can we simplify it?

$$P(B|A=a,E=e) = \alpha\, P(B)P(E=e)P(A=a|B,E=e)$$

Any **hidden variable** that is not an ancestor of a query or evidence variable is irrelevant to the query, and **can be removed before evaluation**.